KL UNIVERSITY · ENGINEERING CAPSTONE PROJECT SPECIFICATION

CLOUD-03 — Predictive Autoscaling with Load Forecasting

Canonical ID
CLOUD-03

Dominant specialization
Cloud Native Software Engineering

Technical domain
Cloud-Native Software Engineering

Strongest impact
Industry Impact + Research Impact

Reference source: keda.sh · KEDA (CNCF)
https://keda.sh/

READ THIS BEFORE CHOOSING THE PROJECT

Two-semester capstone project charter

CP1 + CP2 · complete engineering system

PROBLEM TO SOLVE

Reactive autoscaling always lags demand, over-provisioning to be safe. The consequence is unreliable service behaviour, difficult recovery or avoidable operational cost in Software, Cloud & Developer Infrastructure.

WHO BENEFITS AND WHAT OWNERSHIP IS ESTABLISHED

Primary beneficiaries are software developers and maintainers; the cited source establishes context or source custody but does not identify an operational problem owner. No sponsor, funder or supporting organisation for the KL student implementation is stated. The KL deployment stakeholder and operational decision owner must be confirmed before approval.

FINAL ACHIEVEMENT REQUIRED

Deliver a forecast-driven predictive autoscaler benchmarked against reactive HPA on SLO adherence and cost.

REFERENCE TO REPRODUCE IN CP1

Build a load-forecasting model (Prophet/LSTM) on the traces. This is the O2 reference against which every Predictive Autoscaling with Load Forecasting contribution must be compared.

DIFFERENTIATING CONTRIBUTION IN CP2

Build the predictive scaler (KEDA + forecast) that scales ahead of demand.

COMPLETE FINAL DEMONSTRATION

Run controlled load tests comparing reactive vs predictive scaling. The final demonstration must show the complete Predictive Autoscaling with Load Forecasting workflow rather than an isolated component.

INDEPENDENT ACCEPTANCE

WHY THIS IS CAPSTONE-WORTHY

Evaluate SLO adherence, cost and over-provisioning. The result must be independently checked under representative, boundary and failure conditions.

This is a two-semester culminating project because the team must establish the O2 reference “Load-forecasting model Prophet/LSTM on the traces”, design and evaluate the O3 contribution “Predictive scaler KEDA + forecast that scales ahead”, integrate the O4 system “Controlled load tests comparing reactive vs predictive scaling”, and defend the O5 result under independent, boundary and failure conditions.

Problem evidence and approval condition

OFFICIAL CONTEXTUAL EVIDENCE REQUIRING LOCAL PROBLEM VALIDATION

Evidence locator: Kubernetes event-driven autoscaling + ML forecasting · **Open cited source**

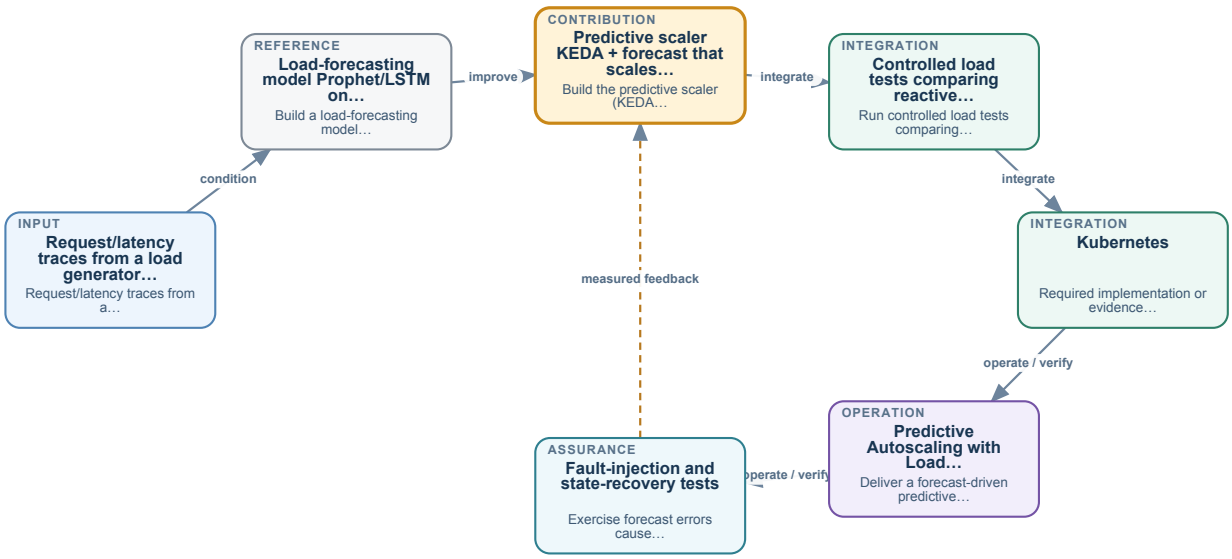
What the source establishes: The cited source establishes the engineering context, technology or application area; it does not by itself prove that the exact local problem remains unresolved.

What the team must verify: Before implementation proceeds, reproduce the stated limitation with versioned evidence or attach a signed problem-validation record from a named stakeholder. If neither is achieved, replace or rescope the project.

Pictorial project specification

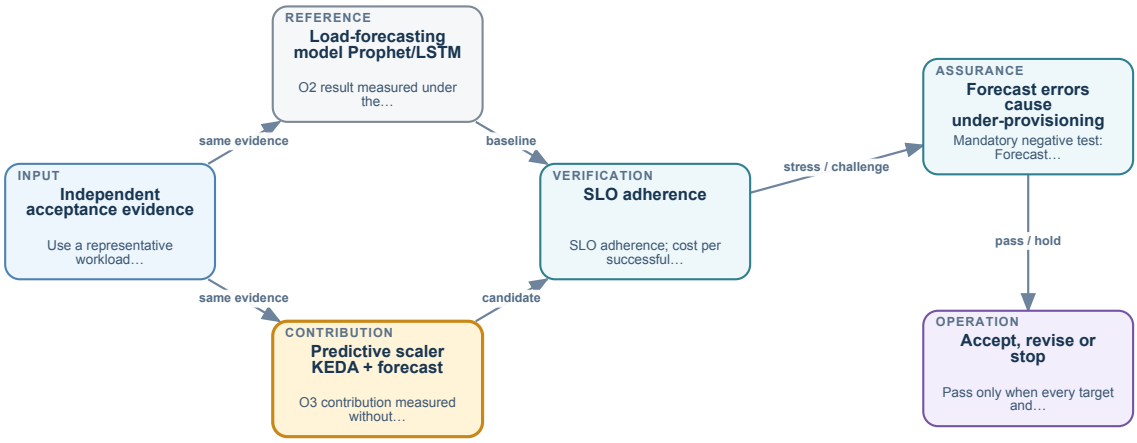
reconciliation loop

A. System / physical architecture — what the team must build and connect



Every arrow is an interface or evidence transfer that must be implemented and tested.

B. Validation architecture — how the team must prove the result



Every arrow is an interface or evidence transfer that must be implemented and tested.

Two-semester execution roadmap

Semester VII: problem + reference · Semester VIII: contribution + system + acceptance

SEMESTER VII / CAPSTONE PROJECT I

O1	**Validate the problem and freeze the engineering contract** Deploy a workload with the default HPA and a load generator; capture request traces. Verify the beneficiary, source claim, system boundary, evidence access, interfaces, hazards and acceptance rules before implementation proceeds. **CP1 Gate 1 — the problem, scope, evidence and acceptance plan are approved.**	**Mandatory outcome and evidence** A baseline HPA setup with recorded load and scaling behaviour. Required evidence: source-to-claim map, stakeholder or reproduction evidence, requirements, architecture decision record, risk register and versioned manifests, workload traces, service telemetry, fault-injection and state-reconciliation evidence.
O2	**Build the reproducible reference and vertical slice** Build a load-forecasting model (Prophet/LSTM) on the traces. Compare at least two feasible reference approaches, select one through a documented trade-off and measure it under AC-1 and AC-4. **CP1 Gate 2 — a reproducible reference and an integrated vertical slice are demonstrated.**	**Mandatory outcome and evidence** A load forecaster with reported accuracy. Deliver executable fixtures, an independent oracle, unit tests, resource profile, failure log and baseline values for SLO adherence; cost per successful transaction; over-provisioned resource-hours per workload unit; forecast accuracy.

SEMESTER VIII / CAPSTONE PROJECT II

O3	**Engineer and prove the differentiating contribution** Build the predictive scaler (KEDA + forecast) that scales ahead of demand. Compare it with O2 using the same evidence and resource envelope; include ablation, sensitivity, trade-off or controlled experiments appropriate to the project. **CP2 Gate 3 — the contribution is technically defensible and its trade-offs are measured.**	**Mandatory outcome and evidence** A predictive autoscaler scaling proactively on forecast load. Report baseline-versus-candidate results, uncertainty, condition-wise failures and a defensible positive or negative conclusion. Retain the selected configuration, experimental controls, raw evidence and a trace from each claim to its acceptance result.
O4	**Integrate the complete operable system** Run controlled load tests comparing reactive vs predictive scaling. Connect the evidence path, reference and contribution, user or operator interface, configuration, security or safety controls, telemetry, degraded mode and recovery. **CP2 Gate 4 — the complete system operates normally, degrades safely and recovers.**	**Mandatory outcome and evidence** Load-test results comparing the two policies. Every component and interface must have unit, integration and system evidence in a representative environment. Retain configuration, interface contracts, telemetry, degraded-mode behaviour and recovery proof.
O5	**Complete independent acceptance and impact validation** Evaluate SLO adherence, cost and over-provisioning. Execute AC-1 to AC-4 and all 5 negative tests; obtain independent domain, stakeholder or technical review. **CP2 Final Gate — every target and guardrail passes, or the project is explicitly held or**	**Mandatory outcome and evidence** A validation report on SLO violations, cost and over-provisioning reduction. Publish baseline, target and achieved values, limitations, residual risks, reproducibility package, operating guide, final report, demonstration video and individual contribution evidence.

revised.

contribution evidence.

Functional requirements

- FR-1 — Ingest or acquire “Request/latency traces from a load generator e.g.”; validate version, provenance, schema or protocol, units and project-specific quality gates before use.
- FR-2 — Reproduce “Load-forecasting model Prophet/LSTM on the traces” through a one-command, one-protocol or one-test-run reference path with an independent expected-result oracle.
- FR-3 — Implement “Predictive scaler KEDA + forecast that scales ahead” as a separately testable subsystem whose configuration and evidence are versioned.
- FR-4 — Implement and verify “Controlled load tests comparing reactive vs predictive scaling” and its input, output, error, configuration and telemetry contracts.
- FR-5 — Implement and verify “Kubernetes” and its input, output, error, configuration and telemetry contracts.
- FR-6 — Deliver “Predictive Autoscaling with Load Forecasting” as the complete demonstrable result; no isolated notebook, model, function or dashboard counts as completion.
- FR-7 — Detect the condition described by NT-1, invoke the documented safe or degraded response and retain recovery evidence.

Non-functional and domain-assurance requirements

- NFR-1 — Performance: meet every accepted KPI target under AC-1 and AC-4, including tail values and resource cost where applicable.
- NFR-2 — Reliability and recovery: pass every mandatory negative test; a critical failure cannot be hidden by an average.
- NFR-3 — Domain assurance: The system shall define SLOs for availability, tail latency, recovery and data integrity under the approved load.
- NFR-4 — Evidence quality: Retries, timeouts, concurrency and partial failures shall not cause silent state loss or duplicate work.
- NFR-5 — Maintainability: use version control, automated tests, documented interfaces, configuration validation and a clean-environment build or protocol.
- NFR-6 — Operability: provide logs, measurements, health or quality indicators, operator guidance, degraded mode and a verified recovery procedure.

System boundary and permitted scope

INCLUDED

- Reproducible reference: Build a load-forecasting model (Prophet/LSTM) on the traces. This is the O2 reference against which every Predictive Autoscaling with Load Forecasting contribution must be compared
- Differentiating contribution: Build the predictive scaler (KEDA + forecast) that scales ahead of demand
- Integrated result: Run controlled load tests comparing reactive vs predictive scaling. The final demonstration must show the complete Predictive Autoscaling with Load Forecasting workflow rather than an isolated component
- Independent acceptance: Evaluate SLO adherence, cost and over-provisioning. The result must be independently checked under representative, boundary and failure conditions

EXPLICITLY EXCLUDED

- out of scope.
- A notebook, isolated model, single-function prototype, dashboard-only demonstration or unvalidated library integration.
- Production certification, universal generalisation and unattended high-stakes operation outside the validated envelope.
- An isolated notebook, model, function, dashboard or library integration without the complete operating and validation workflow.

DEPENDENCY ASSUMPTION

Access to every required dataset, sample, device, tool, reviewer, licence and approval must be confirmed at the project-approval review; unavailable dependencies require an approved scope revision.

Team, prerequisites, tools and feasibility

RECOMMENDED TEAM

A team of 3–4 students must assign named responsibility for workload and failure evidence leadership, control-plane or platform leadership, service integration and observability leadership, reliability and recovery leadership in Predictive Autoscaling with Load Forecasting; each member must own reviewable implementation and evidence.

PREREQUISITES

Kubernetes, forecasting, load testing.

PRINCIPAL TOOLS

- Kubernetes
- KEDA
- Prophet / LSTM
- k6 / Locust
- Prometheus

FACILITY / COMPUTE

An isolated container or cluster testbed with workload generation, telemetry and fault injection; no production dependency is required. Principal access must include Kubernetes, KEDA, Prophet / LSTM, k6 / Locust.

FALLBACK PATH

Use an isolated local cluster or deterministic simulator while preserving workload shape, failure semantics and telemetry. It may reduce scale but must retain O2, O3, O4 and O5.

Acceptance conditions

AC-1	**Representative operation**	Evaluate SLO adherence, cost and over-provisioning.
AC-2	**Boundary and failure operation**	Exercise forecast errors cause under-provisioning; scaling lag and cold starts; synthetic load must resemble real patterns.
AC-3	**Independent acceptance evidence**	Use a representative workload and an independent fault campaign with versioned configuration and workload traces.
AC-4	**Frozen resource envelope**	Use the same approved versions, workload, hardware or facility limits and resource budget for reference and final comparisons; principal tools are Kubernetes, KEDA, Prophet / LSTM, k6 / Locust.

KPI baseline, target, method and pass contract

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-1	**SLO adherence**	O2 reference	higher is better	%	≥ O2 reference	AC-1, AC-3	Use versioned fixtures and an	No critical negative

		measured with the same evidence, workload and resource envelope			+ 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%		independent oracle; report condition-wise values, repeated trials and uncertainty.	test, data-integrity failure or unapproved requirement regression is permitted.
KPI-2	**cost per successful transaction**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-3	**over-provisioned resource-hours per workload unit**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	resource-hours per workload unit	≤ 0.80 × O2 reference per accepted result, with no critical quality regression	AC-1, AC-4	Measure total and per accepted result, including compute, storage, network, material or energy overhead applicable to the system boundary.	The primary correctness, safety or quality metric may not regress by more than the approved non-inferiority margin.
KPI-4	**forecast accuracy**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-5	**successful recovery rate under fault injection (%)**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted.
KPI-6	**service**	O2	lower is	%	≤ 0.80 ×	AC-1, AC-3	Use versioned	No critical

	error rate at declared load (%)	reference measured with the same evidence, workload and resource envelope	better		O2 reference; no critical condition may be worse than the reference		fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	negative test, data-integrity failure or unapproved requirement regression is permitted.

Mandatory negative-test and recovery campaign

NT-1 Forecast errors cause under-provisioning. This condition is exercised within the Predictive Autoscaling with Load Forecasting operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “Forecast errors cause under-provisioning. This condition is exercised within the Predictive Autoscaling with Load Forecasting operating or validation workflow”.

Expected safe behaviour: The service must remain within the approved degraded-mode contract or fail closed, with no silent loss, duplicate work or unreconciled state.

Recovery: Reconcile desired and observed state, replay safely from the last durable checkpoint and prove that no request or state was lost or duplicated.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-2 The cloud or distributed system encounters scaling lag and cold starts during the Predictive Autoscaling with Load Forecasting operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The cloud or distributed system encounters scaling lag and cold starts during the Predictive Autoscaling with Load Forecasting operating or validation workflow”.

Expected safe behaviour: The service must remain within the approved degraded-mode contract or fail closed, with no silent loss, duplicate work or unreconciled state.

Recovery: Reconcile desired and observed state, replay safely from the last durable checkpoint and prove that no request or state was lost or duplicated.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-3 The project violates the requirement that synthetic load must resemble real patterns.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The project violates the requirement that synthetic load must resemble real patterns”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Reconcile desired and observed state, replay safely from the last durable checkpoint and prove that no request or state was lost or duplicated.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-4 The cloud or distributed system encounters A control-plane or dependency outage prevents bounded recovery in Predictive Autoscaling with Load Forecasting during the Predictive Autoscaling with Load Forecasting operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The cloud or distributed system encounters A control-plane or dependency outage prevents bounded recovery in Predictive Autoscaling with Load Forecasting during the Predictive Autoscaling with Load Forecasting operating or validation workflow”.

Expected safe behaviour: The service must remain within the approved degraded-mode contract or fail closed, with no silent loss, duplicate work or unreconciled state.

Recovery: Reconcile desired and observed state, replay safely from the last durable checkpoint and prove that no request or

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or

state was lost or duplicated.

measurements, recovery time and residual-risk decision.

NT-5 **Configuration drift or an unsafe rollout bypasses the declared policy in Predictive Autoscaling with Load Forecasting.**

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “Configuration drift or an unsafe rollout bypasses the declared policy in Predictive Autoscaling with Load Forecasting”.

Expected safe behaviour: The service must remain within the approved degraded-mode contract or fail closed, with no silent loss, duplicate work or unreconciled state.

Recovery: Reconcile desired and observed state, replay safely from the last durable checkpoint and prove that no request or state was lost or duplicated.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

Evidence and reproducibility contract

PRIMARY EVIDENCE

Request/latency traces from a load generator (e.g., k6/Locust)

PROTOCOL AND QUALITY

Version the cited evidence and project inputs, reproduce “Build a load-forecasting model (Prophet/LSTM) on the traces”, implement “Build the predictive scaler (KEDA + forecast) that scales ahead of demand”, and compare both under AC-1 to AC-4 without changing the approved evidence or resource envelope. Check workload representativeness, configuration versions, clock alignment, trace completeness, fault labels and recovered state.

PROVENANCE AND INDEPENDENT PARTITION

The Predictive Autoscaling with Load Forecasting evidence manifest must record source URL, access date, licence or permission, owner/custodian, record/sample/fixture identifiers, exclusions, transformations, configuration and the objective or KPI using each item. Reserve the AC-3 evidence before O3 tuning and separate it by the independent unit appropriate to cloud and platform engineering; no transformation may leak acceptance information into development.

REPRODUCIBILITY

A clean environment must rebuild the O2 reference, O3 contribution, integrated system and acceptance evidence for Predictive Autoscaling with Load Forecasting using the version-pinned toolchain (Kubernetes, KEDA, Prophet / LSTM, k6 / Locust), seeds/configuration and an automated or fully scripted runbook.

Mandatory final deliverables

- D1 — Approved Predictive Autoscaling with Load Forecasting problem charter, requirements, system boundary, architecture and source-to-claim map.
- D2 — Reproducible Load-forecasting model Prophet/LSTM on the traces with fixtures or protocol, oracle, tests, baseline results and resource profile.
- D3 — Independently testable Predictive scaler KEDA + forecast that scales ahead with design rationale, comparative evidence, trade-offs and limitations.
- D4 — Complete Predictive Autoscaling with Load Forecasting prototype or experimental system with configuration, interfaces, telemetry, degraded operation and recovery.
- D5 — Automated or documented acceptance harness covering AC-1 to AC-4, all KPIs and every negative test.
- D6 — Versioned repository or protocol package, evidence manifest, installation/operating guide, API/interface documentation and reproducibility instructions.
- D7 — Final report, research paper or technical article, poster, portfolio case study, demonstration video, video logs and individual contribution/viva evidence.

Key terminology used in this specification

SLO: Service Level Objective — a measurable reliability or performance objective over a defined period.